REVIEW-INFORMED SKINCARE PRODUCT DISCOVERY

Using NLP and Data Analytics

Week 1: Data Science Orientation & Problem Definition

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Problem definition and research plan

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Abstract

The growing use of online platforms for purchasing skincare products has made customer reviews and ratings an important source of information for understanding consumer preferences and product experiences. However, analysing large volumes of customer feedback manually can be time-consuming and may make it difficult to identify recurring opinions and concerns. This project, titled *"Review-Informed Skincare Product Discovery Using NLP and Data Analytics,"* aims to explore how data science techniques can be used to analyse customer reviews and ratings of skincare products.

The proposed approach involves collecting a suitable publicly available dataset, cleaning and preparing the data, and performing exploratory data analysis using Python libraries such as Pandas, NumPy, and Matplotlib. Natural Language Processing (NLP) techniques will be explored to identify common themes in customer feedback, including product texture, fragrance, packaging, perceived effectiveness, and value for money. The analysis will also investigate potential relationships between review topics and customer ratings.

The expected outcome is a structured understanding of customer opinions, recurring product-related concerns, and patterns in review data. These insights may help skincare businesses identify areas for further investigation, improve product information, and support customer-focused decision-making. As this is the initial project proposal, the findings and business implications will be determined only after the dataset has been selected and analysed.

Executive Summary

Online skincare shoppers often compare products using star ratings, descriptions, ingredients, and written reviews. A star rating is useful, but it does not explain why a customer liked or disliked a product. Reviews may mention texture, scent, packaging, value, delivery, or perceived performance. Manually reading large numbers of reviews is time-consuming.

This proposal explores how a beauty and wellness retailer could use ratings and review text to summarize customer feedback across skincare categories. The initial scope is exploratory data analysis, text preparation, sentiment exploration, and review-theme analysis. The results could later inform product-page improvements or a review-informed product-discovery feature if the data supports those uses.

This is a Week 1 planning document: no dataset has been analyzed, no model has been trained, and no business improvement is claimed.

1. Industry Orientation and Business Context

NielsenIQ's 2025 report on Indian beauty discusses ingredient transparency and digital discovery, including the role of ratings and reviews in online beauty purchase decisions [1].

- Its e-beauty report describes the growing importance of online channels and digitally enabled beauty shopping [2].
- McKinsey's 2025 beauty analysis highlights consumer scrutiny of product value and efficacy [3].
- These sources establish the relevance of the topic; they do not prove that review mining will improve sales for a particular retailer.

Imagine a retailer selling cleansers, moisturizers, sunscreens, and serums. One product may have a 4.2-star average from 800 reviews; another may have a 4.5-star average from only 20 reviews. Neither average explains the full customer experience. Written reviews may reveal recurring topics, but reading hundreds of comments manually is difficult.

The analysis will ask what customers commonly mention, whether topics differ between categories, and how lower-rated reviews differ from higher-rated reviews. It will analyze what reviewers say, not make medical judgments or infer that a product causes a skin condition.

2. Business Problem Definition

Problem statement: How can an online skincare retailer organize product ratings and written reviews into understandable summaries of customer feedback across skincare categories?

The immediate problem is not a lack of ratings. It is that ratings compress different experiences into a single number, while written reviews are difficult to summarize consistently at scale. Included: review text, star ratings, product identifiers, product

categories, review dates where available, text preparation, exploratory analysis, sentiment exploration, and review-theme summaries.

2.1 Proposed Analytical Outputs and Success Criteria

The project will produce a structured summary of customer feedback using review ratings and written text. The planned outputs include rating distributions, frequently discussed product attributes, comparisons of review themes across rating groups, and a summary of recurring positive and negative feedback.

The initial success criteria will be:

1. **Data suitability:** Confirm that the selected dataset contains usable review text, ratings, product identifiers, and sufficient skincare product coverage.

2. **Data quality:** Measure missing values, duplicate records, invalid ratings, and product metadata matching.

3. **Analytical usefulness:** Produce interpretable tables and charts showing rating patterns and frequently discussed review themes.

4. **Theme validity:** Inspect representative reviews to verify that the identified themes accurately reflect the original text.

5. **Sentiment evaluation:** If automated sentiment analysis is performed, compare its predictions with a manually checked sample and report appropriate evaluation metrics.

These criteria evaluate the quality and usefulness of the analysis. They do not imply that any results have already been achieved or that business performance will necessarily improve.

2.2 Potential Business Value and ROI Framework

The potential business value of this project is to help a skincare retailer identify recurring customer concerns such as packaging complaints, fragrance concerns, or confusion about product features more efficiently than by manually reading reviews one at a time. These findings could help the retailer decide which product descriptions, customer-support guidance, or product-quality issues deserve further investigation. These are potential uses, not proven benefits; review analysis alone cannot establish that sales or customer retention will increase.

ROI measurement approach

Return on Investment (ROI) can be evaluated only when the business has evidence of both the costs and the resulting benefits. A suitable formula is:

ROI (%) = (Financial benefits − Total costs) ÷ Total costs × 100

For a future ROI assessment, the retailer would need to record analysis and implementation costs (such as staff time, software, and any product or content changes), define measurable outcomes, and compare outcomes over an appropriate period. Possible measures include the frequency of specific complaints, customer-support contacts about the same issue, product-page engagement, repeat purchases, or sales provided reliable data is available. A before-and-after comparison alone may not prove causation; seasonality, promotions, pricing, and other changes should also be considered.

Week 1 limitation: No dataset has been analysed and no business costs, financial benefits, or sales outcomes have been measured. Therefore, this report does not claim a numerical ROI. Any future ROI estimate must be based on documented evidence rather than assumed figures.

3. Aim, Objectives and Deliverables

Aim: Develop a reproducible plan for analyzing skincare reviews so customer feedback can be summarized more clearly than by average ratings alone.

Objectives:

1. Identify a suitable dataset and document its source, licence, date range, fields, and limitations.
2. Inspect review counts, rating distributions, missing values, duplicates, and category coverage.
3. Prepare review text while preserving an unchanged copy of the original.
4. Explore common words and phrases across the dataset and within product categories.
5. Compare review themes across rating bands if sample sizes support it.
6. Test a simple sentiment method against a small manually checked sample, if appropriate.
7. Translate observed patterns into cautious, practical questions for a retailer.
8. Document reproducible steps, assumptions, privacy considerations, and limitations.

Week 1 deliverables: a business problem, research questions and hypotheses, data dictionary and proposed schema, methodology and toolstack, evaluation plan, ethics and limitations, and professional report/repository documentation.

4. Research Questions

- RQ1 — Review themes: Which words, phrases, and topics occur most often in the available skincare reviews?
- RQ2 — Category differences: Do common review topics differ across cleansers, moisturizers, sunscreens, and serums?

- RQ3 — Rating differences: How do topics in lower-rated reviews compare with topics in higher-rated reviews?
- RQ4 — Sentiment usefulness: Does a basic sentiment label agree with the tone of a manually checked sample?
- RQ5 — Business use: Which repeated feedback themes could justify closer review of product descriptions, usage instructions, packaging, or delivery experience?

These are exploratory questions, not assumptions about what the dataset will show.

5. Hypotheses to Investigate

These are hypotheses to test only if the dataset contains the required fields and enough observations; they are not findings.

ID	Hypothesis	Potential comparison	Evaluation Criteria
H1	The frequency of review themes differs across skincare categories.	Theme counts/proportions by category	Compare theme counts and proportions across skincare categories, while reporting the number of reviews in each category.
H2	Lower-rated reviews have a different distribution of common terms or themes than higher-rated reviews.	Theme proportions by rating band	Compare the distribution of themes across predefined lower-rating and higher-rating groups. Report group sizes and avoid interpreting

			associations as causal effects.
H3	A simple sentiment method will not perfectly agree with manual interpretation, particularly for mixed or short reviews.	Agreement on manually checked sample	Compare automated sentiment predictions against a manually reviewed sample. If the sample and labels support it, report precision, recall, F1-score, and a confusion matrix.
H4	Ratings combined with review text provide a richer descriptive summary than average ratings alone.	Compare what each summary communicates	Compare the descriptive information provided by average star ratings alone with information obtained by combining ratings and review-theme summaries. Evaluate whether the combined summary communicates recurring customer concerns more clearly.

A chart that appears to support a hypothesis does not prove it. Group sizes, uncertainty, data quality, and alternative explanations matter.

6. Data Requirements and Proposed Data Architecture

Field	Suggested type	Purpose
review_id	Identifier	Identify review records and check duplicates
product_id	Identifier	Link reviews to products
product_name	Text	Make tables and charts interpretable
product_category	Categorical text	Compare categories
review_text	Text	Main input for review-theme analysis
star_rating	Integer/ordinal	Compare language with rating
review_date	Date, if available	Check time coverage
verified_purchase	Boolean, if available	Optional context, not proof of review accuracy
source_name	Text	Document source and licence

The initial study can use two related tables: Products (product_id, product_name, product_category, optional listed price) and Reviews (review_id, product_id, review_text, star_rating, optional review date and verified-purchase indicator). The product ID links reviews to products. Customer identifiers are not needed for this first research question.

Data source strategy

First ask whether the mentor provides or approves a dataset. Otherwise, find a public dataset with clear documentation and a licence that permits the intended use. Verify that it

contains review text, ratings, and product identifiers; check category coverage, language, date range, missingness, duplicates, and licence terms. A public dataset is not automatically unrestricted. A synthetic dataset may demonstrate code structure but must not be used to claim real-world findings.

6.1 Preliminary Dataset Evidence and Source Selection

One candidate source for this project is the Amazon Reviews'23 dataset, documented by the McAuley Lab. Its All_Beauty category contains approximately 701,500 ratings/reviews, 112,600 products, and 632,000 users according to the published dataset summary. The overall dataset contains 571.54 million reviews across 33 categories, with review activity spanning May 1996 to September 2023.

Source: *https://amazon-reviews-2023.github.io/main.html*

These figures describe the published dataset and have not been independently calculated as part of this project. The All_Beauty category includes beauty products beyond skincare; therefore, its total cannot be treated as the number of skincare reviews.

Before selecting the dataset, the project will inspect product titles, categories, descriptions, available review fields, and metadata coverage. A documented filtering approach will be used to identify relevant skincare products, such as cleansers, moisturizers, sunscreens, and serums. The actual number of usable skincare reviews will be reported only after this filtering and data-quality inspection.

The source will be selected based on relevance to the research questions, availability of review text and ratings, product metadata quality, reproducibility, and applicable access and reuse terms. A smaller, relevant dataset is preferable to a larger dataset that cannot reliably support the stated problem.

7. Analytical Methodology

Phase 1 — Data discovery and quality

Read the documentation and licence; load the data with Pandas; inspect rows, columns, data types, missing reviews, missing ratings, duplicates, unique products, rating ranges, and inconsistent category names. Keep an unchanged copy of the raw data and document cleaning decisions.

Output: dataset suitability note and data-quality summary.

Phase 2 — Exploratory data analysis

Count reviews by category and product; calculate rating counts, mean, median, and distribution by category; plot rating distributions and review volumes; compare review length by rating group; and check whether a few products dominate the dataset.

Output: charts with captions and cautious interpretations.

Phase 3 — Text preparation and theme exploration

Preserve the original review and create a separate cleaned-text field. Standardize case and whitespace, handle obvious markup carefully, and avoid removing negation words such as "not." Extract frequent words and phrases using count-based summaries or TF-IDF. Inspect reviews in context before naming a theme. If sentiment analysis is attempted, compare its output with a manually reviewed sample and report disagreements.

Scikit-learn provides CountVectorizer and TfidfVectorizer for converting text into numerical features [6]. CountVectorizer represents text using word or phrase counts, while TfidfVectorizer assigns weights to terms based on their importance within a collection of documents. These methods support text exploration but do not independently understand context or prove that a product has a defect.

Output: frequent-term tables, category comparisons, and a short theme summary.

Phase 4 — Interpretation and communication

Compare patterns across categories and rating groups only when group sizes are sufficient. Distinguish observed text patterns from explanations requiring further evidence. List practical follow-up questions for product, marketing, or customer-support teams and document limitations and reproducibility steps.

Output: evidence-based summary and next-step recommendations after analysis.

8. Evaluation Plan

- Data quality: report missingness, duplicates, invalid ratings, and category coverage.
- Descriptive validity: show group sizes and ensure charts match underlying counts.
- Theme validity: inspect representative reviews rather than naming themes from word frequency alone.
- Sentiment check: if used, manually label a small documented sample and report agreement, disagreements, and failure examples.
- Reproducibility: record library versions, cleaning rules, source details, and notebook steps.
- No unsupported performance claims: do not state accuracy, ROI, churn reduction, or sales uplift before an appropriate test.

9. Python Tool stack

Tool	Planned role
Python	Main analysis language
Pandas	Read, inspect, clean, group data
NumPy	Numerical operations where useful
Matplotlib / Seaborn	Rating and category charts
scikit-learn	Count-based text features and TF-IDF; optional simple models
NLTK or another text utility	Optional text preprocessing
Jupyter Notebook	Keep code, charts, and interpretation together
python-docx	Optional report generation

The project starts with descriptive analysis. Additional libraries will be added only when they are needed.

10. Ethics, Privacy and Limitations

- Use only datasets whose terms permit the intended analysis.
- Avoid collecting names, emails, account IDs, or unnecessary personal information.
- Do not infer a reviewer's skin type or medical condition from text.
- Reviewers are self-selected and may not represent all customers.
- Ratings can reflect delivery, packaging, expectations, or other factors.
- Sarcasm, mixed sentiment, spelling errors, multilingual text, and short reviews can mislead simple NLP methods.
- Frequent terms are not automatically important themes; sentiment labels do not establish product safety or clinical effectiveness.

- Findings from one dataset may not generalize to the whole beauty market.

10.1 Practical Challenges and Mitigation

1. **Product classification:** A general beauty dataset may include makeup, haircare, and other personal-care products. Product titles and metadata will be inspected, and the skincare inclusion rules will be documented.

2. **Incomplete records:** Missing review text, ratings, or product metadata may reduce the usable sample. Missingness and metadata matching rates will be measured before deciding which records can support each analysis.

3. **Review bias:** Customers who submit online reviews may not represent all purchasers. Results will therefore be described as patterns in the available reviews rather than conclusions about every skincare consumer.

4. **Sentiment ambiguity:** Sarcasm, mixed opinions, negation, spelling errors, and short reviews may reduce automated sentiment reliability. Representative predictions and errors will be manually inspected.

5. **Computational constraints:** Large datasets can require substantial storage and processing time. Initial checks will use a manageable sample before processing a larger dataset.

6. **Responsible data use:** The source documentation and reuse terms will be reviewed before collection or redistribution. Unnecessary customer identifiers will not be included in the analytical dataset.

These mitigations are planned procedures. Their effectiveness will be assessed during implementation.

11. Proposed Work Plan

Work block	Activity
1	Read task brief and industry sources; finalize question
2	Identify and verify a suitable dataset
3	Document fields, quality issues, and cleaning rules
4	Plan and create initial EDA summaries and charts
5	Explore text features and validate any sentiment method
6	Interpret evidence, write limitations, and polish report

12. Conclusion and Next Step

This proposal focuses on an achievable first investigation: turning skincare review text and ratings into clearer summaries of customer feedback. It is narrower than a full recommendation or churn platform, making its questions and data requirements easier to verify. The next step is to confirm the topic and dataset with the mentor. Once a suitable dataset is selected, the project can move from a conceptual plan to actual Python analysis.

13. References

1. NielsenIQ. (2025, June 26). *The New Face of Indian Beauty.* https://nielseniq.com/global/en/insights/report/2025/the-new-face-of-indian-beauty/
2. NielsenIQ. (2025, June 26). *The Digital Makeover: E-Beauty Trends in India.* https://nielseniq.com/global/en/insights/report/2025/the-digital-makeover/
3. McKinsey & Company. (2025, June 9). *State of Beauty 2025: Solving a Shifting Growth Puzzle.* https://www.mckinsey.com/industries/consumer-packaged-goods/our-insights/state-of-beauty-2025
4. McAuley Lab. *Amazon Reviews'23 Dataset Documentation.* https://amazon-reviews-2023.github.io/main.html
5. McAuley Lab. *Amazon Reviews'23 Dataset Card.* Hugging Face. https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023
6. scikit-learn developers. *Feature Extraction — Working with Text Data.* https://scikit-learn.org/stable/modules/feature_extraction.html
7. Yadav, N. B. (2023). *Harnessing Customer Feedback for Product Recommendations: An Aspect-Level Sentiment Analysis Framework.* Human-Centric Intelligent Systems, 3, 57–67. https://doi.org/10.1007/s44230-023-00018-2